

Soft Actuators

1. Electrically responsive actuation: Dielectric Elastomer Actuators (DEAs)

Based on Coulombic attraction between two flexible electrodes with a potential difference located on separate ends of a compressible membrane.

Very flexible, high-energy density, high strains, exhibit self-sensing characteristics, have accurate simulation models, and have been demonstrated to be as useful as biological muscles.

Performance: in function to the steadiness, breakdown voltage, dielectric constant of the elastomer (ability to store E energy)

☹: large voltages required to operate, leakage currents, easier to reach electrical breakdown

Materials:

Acrylic elastomers: VHB4910 elastomer, achieve quite large deformations ☹ enormously viscoelastic, limit the actuators bandwidth

Si-based and PU-based elastomers show faster reaction and have the ability to be cast into various shapes and softness ☹ produced strain is significantly lower

Si-based elastomers: reduced permittivity: higher required activation voltage as compared with PUs acrylic elastomers.

Solutions to increase the dielectric constant of the membrane: embedding microparticles/ developing completely organic elastomers

DEAs can: Stacking, folding, framing, inflating, rolling

They can: bent, expanded or buckled, increase application range

e.g., annelid-like robot

Search how does Dielectric elastomer works

Materials: Ecoflex 00-50

Link:

- [Advancements and challenges in dielectric elastomer actuator-based biomimetic mobile robots](#)
- [Progress, Challenges, and Prospects of Soft Robotics for Space Applications](#)
- [Soft Robotics: Examples, Research and Applications](#)
- [Soft Actuators for Soft Robotic Applications: A Review](#)
- [Viscoelastic Material](#)
- [Robots in Disaster Response: Innovations in Crisis Management](#)
- [Self-folded soft robotic structures with controllable joints](#)
- [Magnetic soft robots: Design, actuation, and function](#)

- [Origami-inspired folding assembly of dielectric elastomers for programmable soft robots](#)
- [Locomotion of an untethered, worm-inspired soft robot driven by a shape-memory alloy skeleton](#)
- [Molding and Casting for Soft Robotics](#)
- [Light-driven soft microrobots based on hydrogels and LCEs: development and prospects](#)
- [Recent advances in biomimetic soft robotics: fabrication approaches, driven strategies and applications](#)
- [Soft-gripper](#)
- [Earthworm robots could help astronomers explore other worlds](#)
- [softroboticstoolkit](#)

Embodying soft robots with octopus-inspired hierarchical suction intelligence

Hierarchical – arranged in order of rank

Neuromuscular – relationship between nerves and muscles

Dexterous – having the ability to perform a difficult action quickly and skillfully with the hands

In-arm embodied computation – computational processes are physically integrated into the structure of a robotic arm or limb, enabling more efficient, adaptive, and intelligent movement and interaction with the environment.

Local fluidic circuitry – use of fluid-based (liquid or gas) systems to perform computation, control or signal processing directly within a device or structure, like soft robot, hydraulic machine, microfluidic chip. Fluidic circuits manipulate flow, pressure and fluid properties to achieve logical operations, mechanical actuation, or autonomous control, without relying on external electronics often. This is different from electronic circuit

Devolved: having had power transferred or delegated to a lower level, especially from central government to local or regional administration

Ganglion: a group of neuron cell bodies in the peripheral nervous system (-> components make up the nervous system of bilateral animals (-> a large clade of animals characterized by bilateral symmetry (-> contain a single plane of symmetry, the sagittal plane, which divides the organism into two roughly mirror image left and right halves))) during embryonic development (-> developmental stage of an animal embryo (-> initial stage of development for a multicellular organism. In organisms that reproduce sexually, embryonic development is a part of the life cycle that begins just after fertilization of the female egg cell by the male sperm cell)), with the other part being the central nervous system (CNS) (-> the system control brain, retina and spinal cord)

Brachial: adj that means of, relating to, or situated in the arm or an armlike process. It is often used in anatomy to describe parts of the arm, such as the brachial artery. It can also refer to the foreleg, wing, pectoral fin, or other forelimb of a vertebrate

So, brachial ganglion: the peripheral nervous system in arm muscle

Rim: the upper or outer edge of an object, typically something circular or approximately circular

Receptor: a cell or group of cells that receives stimuli

Adhesion: the action or process of adhering to a surface or object

Cord: a length of rope or string made of twisted threads

Bandwidth: maximum rate of data transfer across a given path

Tactile: of or connected with the sense of touch

Proprioception: ability to sense your own body position and movement

Paper demonstration:

By incorporating simple suction cups into fluidic-driven soft circuits, robots can establish octopus – like hierarchical and distributed neuromuscular system and mimic the integrated suction adhesion, embodied intelligence, and multimodal perception of the octopus

Experiments:

1. Introduce a suction-based embodied intelligence strategy based on the local integration of sensing and control within the robot body, which enables it to achieve octopus-like low-level autonomy, including grasping weakly supported delicate objects and sucking a curling around unknown object.
2. A suction-based multimodal sensing strategy, which is achieved by a simple signal classifier and enables the robot to perceive multiple environmental conditions, including contact, fluidic viscosity, surface roughness, and pulling forces.
3. The fusion of the proposed suction-based embodied intelligence and multimodal perception strategies and thereby achieve integrated adhesion, embodied intelligence, and multimodal perception in a simple suction gripper with a hierarchical neural structure. Once it receives an activation signal from the higher-level controller, the gripper can autonomously execute the extend-suck-retract action while simultaneously detecting and measuring object properties.'

Embodied: include or contain (sth) as a constituent part

Efferent (adj): conducted or conducting outwards or away from something (for nerves, the central nervous system; for blood vessels, the organ supplied).

Afferent: opposite of efferent

Perceptual: relating to the ability to interpret or become aware of something through the senses

Distal: situated away from the centre of the body or from the point of attachment. The opposite of [proximal](#).